Giovanni Guidarini

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Education

MSc in Computational Science and Engineering Sep. 2025 – Present
ETH Zurich University *Zurich, Switzerland*

- Specialization: Computational Physics
- Relevant coursework: HPC Lab; Large-Scale Convex Optimization; Probabilistic Artificial Intelligence; AI in the sciences and Engineering; Quantum Information Processing; Quantum Information Theory; Programming techniques; Computational Statistical Physics.
  Ongoing coursework: Embedded Systems; HDL-Based Design for FPGAs; Discrete-time Signal Processing; Quantum Error Correction.

BSc in Mathematical Engineering Sep. 2022 – Jul. 2025
Politecnico di Milano University *Milan, Italy*

- Graduated 110/110 with Honors
- Relevant coursework: Quantum Mechanics, Probability Theory, Statistical Inference, Numerical Analysis, Linear Algebra, Partial Differential Equations, Functional and Complex Analysis
- Member of AIM – Association of Mathematical Engineers

Projects

Particle Track Reconstruction as a Spin-Glass Optimization Problem | *Monte-Carlo, C++, Python*
Computational Statistical Physics (FS 2026) Spring 2026

- Mapped LHC particle track reconstruction onto an Ising-like spin-glass model with a physics-motivated Hamiltonian; built a C++ simulated-annealing solver minimizing a three-term energy function encoding alignment, length, and curvature constraints.
- Tuned hyperparameters with Optuna on the Euler cluster, reaching 85% strict track efficiency on clean events.

AI in the Sciences and Engineering | *Deep Learning, PyTorch, Scientific Machine Learning* Fall 2025

- Investigated pathological optimization in Physics-Informed Neural Networks (PINNs) on a multiscale Poisson equation, using Hessian-eigenvector analysis to link high-frequency components to ill-conditioned loss landscapes and training instability.
- Trained a Fourier Neural Operator as a neural surrogate for an unknown dynamical system; evaluated resolution invariance and demonstrated transfer learning to out-of-distribution regimes. Extended the Geometry-Aware Operator Transformer (GAOT) with KNN-based radius aggregation for irregular geometries.

Generative Antibody Design with Protein Language Models | *Deep Learning, PyTorch, PLMs*
Learning & Adaptive Systems Group (Prof. Andreas Krause, Scott Sussex) Spring 2026

- Turned ESM2, a masked protein language model, into a generative model for antibody design via beam search over the CDRH3 region, proposing high-affinity variants instead of random mutagenesis.
- Fine-tuned ESM2 via Direct Preference Optimization on binding-affinity data; built the end-to-end training/evaluation pipeline (Hydra, Slurm) comparing vanilla, evotuned, and DPO-tuned models.

Opinion Dynamics in Social Networks with Community Structure | *Modeling, Simulation* Spring 2025

- Experimental BSc thesis (supervised by Prof. C. Piccardi) on how community structure affects opinion spread and manipulation, via mathematical modeling of opinion diffusion and simulation over large networks.

Experience

Artificial Intelligence team intern Jul. 2024
Anima SGR *Milan, Italy*

- Independently built and backtested a sentiment-driven sector rotation strategy on STOXX 600 equities (2012–2024) using RavenPack data; best strategy achieved 0.50 Sharpe vs. 0.31 benchmark. Presented to Head of Quant Division and CFO; strategy was later adopted into the firm's production pipeline.

Technical Skills

Programming: C/C++, MPI, OpenMP, Python, CUDA, Fortran, PyTorch, Git, Linux, Slurm

Additional Information

Languages: Italian (Native), English (IELTS 8 - C1)
Interests: Motorsports, AI, Art, History and Literature, former competitive rugby player